Unexpected Online Financial News and Intraday Market Reactions: An Event-Study of EUR/USD and Nasdaq-100 Prices

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Abstract

We study the intraday response of EUR/USD and the Nasdaq-100 index to 2,449 unscheduled news events from a public real-time newsfeed (FinancialJuice Discord) over a thirteen-month sample from 24 March 2025 to 5 May 2026, with each event independently labelled by a large language model (`deepseek-v4-flash`) and benchmarked against the Loughran-McDonald finance dictionary and the FinBERT-tone neural classifier. Using a matched, hour-of-day and day-of-week stratified baseline, with cluster-robust standard errors and uniform Benjamini-Hochberg false discovery rate adjustment, we find that news events are robustly associated with intraday magnitude that is 1.3× to 1.9× the baseline level on every asset-window cell, and that the intra-window range and maximum absolute move are the most stable single results. The directional content of LLM sentiment is modest (49–54% hit rate) and below transaction costs in a backtest, but the LLM outperforms both baselines by 2–6 percentage points on the primary windows and is the only classifier that produces asset-specific sentiment, doing so on 73.5% of events. On the Financial PhraseBank external benchmark, the LLM achieves 84.6% accuracy and Cohen's $\kappa = 0.72$ against gold human labels, outperforming FinBERT by 5.4 percentage points and the Loughran-McDonald dictionary by 27.2 percentage points. The pre-event absolute return is comparable in magnitude to the post-event return, indicating that the Discord publication time is not the underlying informational event time and that entry at the headline timestamp is, on average, already late. The initial price reaction extends rather than reverses over the subsequent hours. The full pipeline, including the LLM cache, the dictionary and FinBERT baselines, the matched-baseline construction, and the validation harness, is released as a single seeded reproducible repository.

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JEL classification: G14, G15, C58, C45.

1 Introduction

Information enters financial markets through a heterogeneous network of public news services, professional terminals, broker desks, and, in the last decade, real-time social and chat-based aggregators that an increasing share of retail and small-shop traders monitor as their primary news feed. These newer channels carry the same headlines that appear on Bloomberg or Reuters but at a variable latency and with a publication timestamp that is not the timestamp of the underlying primary source. Whether and how prices react to the arrival of news on such channels—and whether the directional content of those headlines can be extracted in a useful way by modern language models—is a question with direct practical relevance to a large and growing audience of intraday traders, and a methodological question for the event-study literature that has historically focused either on scheduled macro releases (Andersen et al. 2003) or on slower-moving newspaper text (Tetlock 2007; Loughran and McDonald 2011).

This paper studies the relation between unscheduled news headlines from a representative real-time news service—the FinancialJuice Discord newsfeed—and intraday price reactions in the EUR/USD spot exchange rate and the Nasdaq-100 index, over a thirteen-month sample from 24 March 2025 to 5 May 2026. The dataset comprises 2,449 hand-filtered "gold" news events drawn from 63,016 raw messages and aligned to one-minute price bars on both assets. Each event is independently labelled by a large language model with a discrete sentiment for USD and for NDX, an expected magnitude, a surprise level, and a self-reported confidence. We test fourteen pre-specified hypotheses covering volatility, direction, persistence, time-of-day, cross-asset behaviour, and pre-event drift, and we report the central findings with Benjamini-Hochberg false discovery rate adjustment (Benjamini and Hochberg 1995) applied uniformly across the full hypothesis battery.

Three findings warrant emphasis. First, the news events are robustly associated with elevated intraday magnitude: the mean absolute return, the high-low range, and the maximum absolute intra-window move are between 1.3× and 1.9× their matched-baseline counterparts across all assets and windows tested, and the result is robust to winsorisation at the 1% tails, to the introduction of multivariate controls for category, surprise, magnitude, confidence, cluster size, and pre-event drift, and to a pre/post split at the LLM's training data cutoff. The intra-window range and maximum-absolute-move statistics are, in fact, the strongest and most stable result in the analysis. Second, the directional content of LLM-derived sentiment is modest: direction hit rates range from 49% to 54% on the primary windows, with only a small number of cells surviving the FDR correction. A backtest with realistic spread and slippage costs confirms that this edge is too small to be economically exploitable in the naive forms tested. Third, the LLM sentiment nonetheless out-performs both the standard Loughran-McDonald finance dictionary (Loughran and McDonald 2011) and the FinBERT-tone neural classifier (Yang et al. 2020) on direction hit rate by 2–6 percentage points across the primary windows, and produces cross-asset disagreement (different sentiment for USD and for NDX on the same event) on 73.5% of events—a property that neither single-polarity baseline can express by construction.

We also report two methodologically substantive findings. The absolute return in the fifteen minutes preceding a Discord headline is elevated above the matched baseline by a similar magnitude as in the post-event window. Without primary-source timestamps we cannot distinguish whether this reflects information arrival latency in the public feed or genuine pre-disclosure activity, but the practical implication for users of the feed is the same: entering a trade at the timestamp of the public headline is, on average, already late. And the sign of the +15-minute return agrees with the sign of the +4-hour return on 57.6% of events, with median absolute moves 3–4.5× larger over the longer window, so the initial reaction tends to be extended rather than reversed.

Our contribution is fourfold. We assemble a sizeable intraday event-study dataset on an actively-used public news feed and document its construction in enough detail to support replication. We propose an event-study workflow with matched-baseline standardisation, explicit cluster dependence handling, and uniform FDR adjustment as applied to a large battery of hypotheses. We compare an LLM-based sentiment classifier to two reference baselines—the Loughran-McDonald finance dictionary and the FinBERT-tone neural classifier—on identical event windows, providing direct quantitative support for the LLM methodology against both classical and BERT-era baselines. And we make the entire pipeline—including the validation harness, the LLM cache, the dictionary baseline, the FinBERT baseline, and the three-way comparison—public in a single repository with seeded reproducibility (see Reproducibility Statement).

The remainder of the paper is organised as follows. Section 2 reviews the relevant literature on event studies, financial sentiment analysis, and online news. Section 3 describes the data. Section 4 develops the methodology. Section 5 presents the results, grouped by theme. Section 6 discusses the implications. Section 7 details the limitations of the design and the directions for future work, and Section 8 concludes.

2 Literature Review

Our work intersects four established literatures. We organise the review around those four threads, taking care to identify, for each, the specific result that motivates a methodological choice in our own design or that frames the contribution of our findings.

2.1 Event studies and intraday price discovery

The event-study method, formalised in its modern shape by MacKinlay (1997), exploits the proposition that, under semi-strong market efficiency (Fama 1970), the price response to an informational event reveals its market-relevant content. The classical approach computes abnormal returns relative to a market model on daily data; the high-frequency extension reduces the observation horizon to intraday windows of seconds or minutes.

The canonical high-frequency study on macro news is Andersen et al. (2003), which uses six years of real-time foreign-exchange quotes to characterise the conditional mean response of major USD pairs to scheduled macroeconomic announcements. Two features of that literature inform our design. First, the appropriate benchmark in the high-frequency setting is a matched non-event window of the same length, not a long-horizon market model, because intraday volatility is dominated by hour-of-day and day-of-week seasonality that would otherwise contaminate the abnormal-return estimate. Second, on FX, USD strength is a derived quantity: a study that uses the EUR/USD price directly to test a USD-oriented hypothesis (e.g., "good news for the dollar produces a positive reaction") will assign the wrong sign to every event whose target asset is the dollar rather than the euro. We adopt both conventions: a matched, hour-of-day and day-of-week stratified baseline (Section 4.4) and an explicit USD proxy convention $r^{\mathrm{USD}} = -r^{\mathrm{EURUSD}}$ (Section 4.2).

Most of the existing high-frequency event-study literature focuses on scheduled macro releases—CPI, non-farm payrolls, central-bank decisions—because the release time is known in advance and clean event windows are straightforward to construct. Unscheduled news, such as geopolitical escalations or political statements, has received comparatively less attention at the intraday horizon. Our sample is dominated by such unscheduled events (we deliberately filter out the publisher's `$MACRO` tag, which marks scheduled releases), placing the analysis closer to the spirit of the news-text literature than to the macro-announcement tradition.

2.2 Financial sentiment analysis: from dictionaries to LLMs

The textual analysis of financial news as a predictor of returns begins with Tetlock (2007), who shows that the fraction of negatively-valenced words in the Wall Street Journal "Abreast of the Market" column predicts next-day index returns and reverses over the following week. Tetlock uses the Harvard IV-4 General Inquirer dictionary, an inheritance from political-science content analysis.

Loughran and McDonald (2011) subsequently establish that general-purpose sentiment dictionaries misclassify a large share of words in financial text—the canonical example is "liability," which is financially neutral but Harvard-negative—and they construct a finance-specific dictionary that has since become the standard baseline in the field. We use the Loughran and McDonald (2011) dictionary explicitly as our reference baseline against which the LLM-derived sentiment must improve in order to justify the methodological choice of an LLM (Section 5.2). We do not use Harvard IV-4, partly because Loughran and McDonald (2011) themselves demonstrate its inadequacy in this context.

Antweiler and Frank (2004) and Bollen et al. (2011) extend the textual-analysis program from professional newspaper text to user-generated content (stock message boards and Twitter respectively), demonstrating in both settings that simple bullishness counts have non-trivial predictive content for return and volatility, albeit small. Our study is in this tradition in that the FinancialJuice Discord feed is a real-time professional-curated aggregator broadcast to a primarily retail audience, rather than a peer-reviewed wire service.

The modern wave of work uses neural language models. Heston and Sinha (2017) benchmark a Thomson-Reuters proprietary neural network against Harvard and Loughran-McDonald on a large news corpus and find that the neural model captures both faster and more persistent reactions, particularly for negative stories. Yang et al. (2020) release FinBERT, a BERT-family model pre-trained on a 4.9-billion-token corpus of financial communications, with state-of-the-art results on the Financial PhraseBank benchmark. Calomiris and Mamaysky (2019) combine topic modelling with sentiment and unusualness ("entropy") signals on a multi-country Reuters corpus and show predictive content for country-level returns and volatilities at horizons up to one year. We extend this line of work to a contemporary large-language-model classifier (`deepseek-v4-flash`) applied to a real-time chat-aggregated news feed, with explicit benchmarking against the dictionary baseline and a self-consistency check against a more capable sibling model.

2.3 Online news, sentiment, and intraday reactions

The intersection of social or online news with intraday market behaviour has been studied at the daily horizon by Bollen et al. (2011) (mood from Twitter predicts the Dow Jones at several days' lead) and at the search-volume level by Da et al. (2015) (aggregated Google Trends queries for words like "recession" and "bankruptcy" predict short-term return reversals and volatility increases). Antweiler and Frank (2004) show that the volume rather than the direction of message-board posts predicts realised volatility—a result we revisit in Section 5.6 with our log(cluster size) control, where larger news bursts produce larger magnitude reactions even after controlling for the LLM sentiment and category labels.

The relatively under-explored corner of this map is the sub-fifteen-minute response to unscheduled news on a real-time public chat feed, which is precisely the setting our dataset addresses.

2.4 Multiple testing and reproducibility

A practical concern when running a large pre-specified hypothesis battery is that the number of tested cells inflates the family-wise type-I error far beyond the nominal level. We follow the Benjamini-Hochberg procedure (Benjamini and Hochberg 1995) and report q-values that control the false discovery rate across the union of all p-values produced by the pipeline. The procedure is more appropriate than the Bonferroni correction for a large hypothesis family in which a non-trivial fraction of nulls are expected to be false. All conclusions in Section 5 are drawn from q-values, with raw p-values reported for transparency. The full pipeline, the seed, and the LLM cache are documented in the Reproducibility Statement to support direct replication.

3 Data

3.1 News data

We use the public newsfeed of the FinancialJuice Discord server, captured via Discord-ChatExporter into a single JSON archive. The export covers the period from 24 March 2025 to 5 May 2026 (approximately thirteen months) and contains 63,016 individual messages from the official `FinancialJuice` account. The feed is a real-time aggregation of macroeconomic releases, central bank communication, geopolitical headlines, political statements, energy market news, and corporate announcements, intended primarily for an audience of intraday traders. All timestamps are recorded by Discord at the moment the message is published and converted to UTC.

To isolate market-relevant unscheduled events from routine data releases, we apply a deterministic "gold" filter: a message is gold if (i) it is flagged with a red dot or begins with BREAKING, and (ii) it is not flagged with the `$MACRO` tag that the publisher attaches to scheduled macro indicator releases (CPI, NFP, PMI, etc.). The first condition selects headlines that the publisher itself considers high-impact; the second removes scheduled releases that would otherwise dominate the sample and whose timing carries no informational surprise. The filter retains 2,449 messages (3.9% of the raw feed), which we treat as the working population of unscheduled news events.

We assign each gold event to one of seven categories using deterministic keyword rules embedded in `parse_fj_discord.py`: `macro_release` (residual data releases not caught by the `$MACRO` tag), `central_bank` (Federal Reserve, ECB, Bank of England, etc.), `geopolitical` (war, sanctions, diplomacy), `politics` (executive and legislative statements), `energy` (OPEC, oil inventories, gas), `corporate` (named firm news), and `other`. The category labels are used only for stratified hypothesis testing in later sections and play no role in event selection.

3.2 Sentiment labels

Each gold event is independently labelled by a large language model with respect to its short-horizon directional implication for the U.S. Dollar (USD) and the Nasdaq-100 Index (NDX). We use `deepseek-v4-flash` accessed via an OpenAI-compatible API endpoint. The system prompt (reproduced in the Online Appendix) restricts the model to a strict JSON schema with the following fields per event: a discrete USD sentiment in {`bull`, `bear`, `neutral`}, a discrete NDX sentiment in the same support, signed directional strengths in $[-1, +1]$ for each asset, an expected magnitude in {`low`, `med`, `high`}, a surprise level in {`expected`, `surprise`, `shock`}, a self-reported confidence in $[0, 1]$, and a one-sentence rationale. The prompt explicitly instructs the model to (a) base its judgment only on the supplied text, (b) return `neutral` with low confidence when the message is ambiguous, and (c) treat USD and NDX independently so that risk-off events can map to `bull USD, bear NDX` simultaneously. Five few-shot examples covering geopolitical, central bank, corporate, and neutral cases are included verbatim in the system prompt to anchor the output format. Calls are cached locally in SQLite, keyed on the hash of the model name, system prompt, and event content, so that re-runs during prompt iteration do not duplicate API costs.

The LLM-derived labels are the primary sentiment input throughout the paper. In Section 5.2 we benchmark them against a deterministic Loughran-McDonald dictionary baseline (Loughran and McDonald 2011) and against the FinBERT-tone neural classifier (Yang et al. 2020).

3.3 Price data

We use one-minute open-high-low-close-volume (OHLCV) bars for two target assets from Dukascopy: the EUR/USD spot exchange rate and the `E_NQ-100` Nasdaq-100 contract for difference (CFD). The CFD price track is used as a tradable proxy for the cash Nasdaq-100 index in the absence of an authoritative consolidated tape at one-minute resolution. We acknowledge the proxy nature of this series in Section 7 and discuss its implications for the reported volume tests.

The price coverage spans 24 March 2025 to 5 May 2026, identical to the news coverage window. After de-duplication and gap detection, the EUR/USD panel contains 416,500 bars and the NDX panel contains 381,122 bars; the difference reflects the venue's CFD market hours (no weekend trading, exchange holidays). Closed-period gaps are detected automatically and events that fall inside them are diverted to the closed-period gap test (Section 5.5) rather than to the main intraday windows. A reported volume series accompanies each bar; for the NDX CFD the series is best understood as tick volume rather than consolidated exchange volume, and any volume-based test must be interpreted accordingly.

3.4 Final analysis sample

The two coverage windows (news and prices) are identical, so the intersection used for the event study is also 24 March 2025 to 5 May 2026, and no gold events are dropped for falling outside the price range. After cluster assignment at a 15-minute gap threshold, the 2,449 events organise into 1,405 distinct news clusters; the median cluster contains a single headline, but the largest contain dozens of related releases issued within minutes of each other (typically central bank decisions followed by press conferences, or geopolitical escalations followed by official reactions). Cluster identity is preserved throughout the analysis: non-regression tests use the first event per cluster as an independent observation, and regression standard errors are clustered on cluster identity.

Generating one event-asset-window record for each combination of (event, asset in {EUR/USD, NDX}, window in {1, 5, 15, 60, 240} minutes) yields a panel of 24,490 event-window rows. A parallel cluster-level panel constructed from the first timestamp of each cluster yields 14,050 rows. Both panels are written to the project repository (`event_study_windows.csv` and `cluster_event_study_windows.csv`). Table 1 reports the key sample quantities, Table 2 defines the variables used throughout the analysis, and Table 3 provides descriptive statistics for the key event-level and window-level quantities. Figure 1 plots the arrival rate of gold events by category over the sample period; the rate is fairly uniform across months, with central-bank and political news dominating the mix.

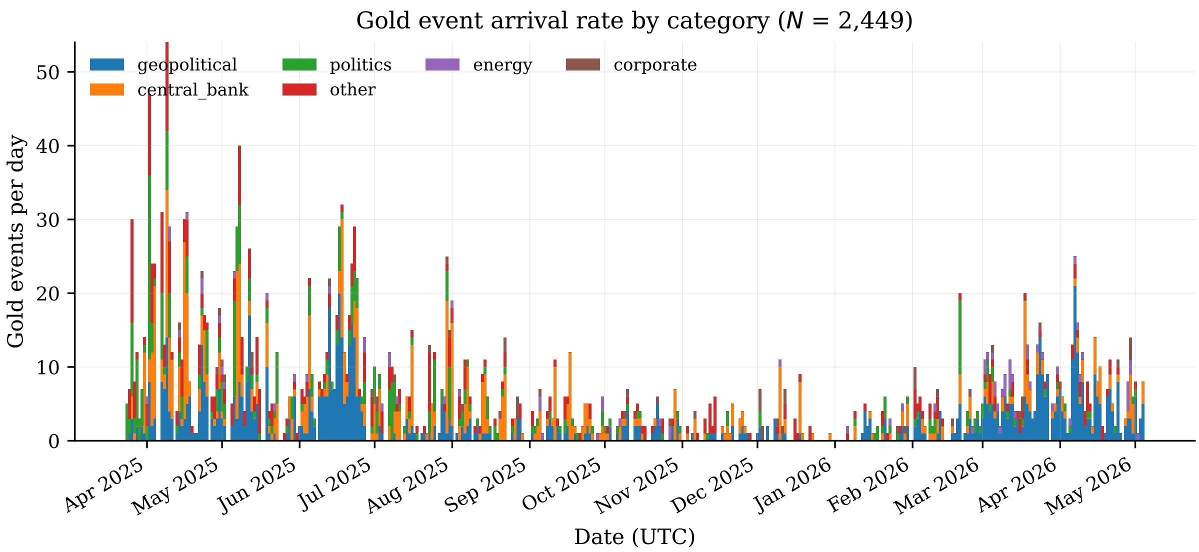

Figure 1: Daily arrival rate of gold news events by category, with the publisher's `$MACRO` tag excluded. The horizontal axis spans the thirteen-month sample window. Stack ordering follows the overall category volume.

Table 4 reports the distribution of the three sentiment classifiers (the LLM, the Loughran-McDonald dictionary, and FinBERT-tone) across the full event sample, together with pairwise label agreement. The three classifiers disagree substantially: pairwise USD agreement is below 40% for every pair, and the LLM is the only one of the three that

Table 1: Sample summary.

Quantity	**Value**
News coverage (UTC)	24 Mar 2025 – 5 May 2026
Raw Discord messages parsed	63,016
Gold events after filter	2,449
Gold rate (gold / raw)	3.9%
Events dropped (outside common price range)	0
News clusters at 15-minute gap	1,405
Median cluster size (events)	1
Event-window panel rows	24,490
Cluster-window panel rows	14,050
EUR/USD 1-minute bars	416,500
EUR/USD coverage	24 Mar 2025 – 5 May 2026
Nasdaq-100 CFD 1-minute bars	381,122
Nasdaq-100 CFD coverage	24 Mar 2025 – 5 May 2026
LLM sentiment model	`deepseek-v4-flash`
Dictionary baseline	Loughran-McDonald (2011)
Neural baseline	FinBERT-tone (Yang et al., 2020)

Notes. Sample drawn from the FinancialJuice Discord newsfeed and aligned to Dukascopy 1-minute OHLCV bars for both target assets. Gold filter: red-dot or BREAKING flag and not tagged `$MACRO`. Cluster identifier groups events separated by no more than 15 minutes. Forward window lengths used throughout: 1, 5, 15, 60, 240 minutes; primary windows reported in Section 5 are 5, 15, and 60 minutes. All timestamps are UTC.

Table 2: Variable definitions.

Variable	Definition	Unit
$r^{\text{price}}_{i,a,W}$	Close-to-close percentage return of asset a over the W-minute post-event window starting at $\lceil t_i \rceil_{1\text{m}}$.	%
$r^{\text{target}}_{i,a,W}$	Target-aligned return. Equal to r^{price} for NDX; equal to $-r^{\text{price}}$ for EUR/USD (USD proxy convention).	%
$\text{range}_{i,a,W}$	Intra-window high minus low, expressed as a percentage of the open price.	%
$\text{maxabs}_{i,a,W}$	Maximum of the absolute upward and downward intra-window excursions, as a percentage of open.	%
$z_{i,a,W}$	Standardised abnormal score: outcome minus matched-baseline mean, divided by matched-baseline standard deviation.	dimensionless
`sentiment_usd`	LLM-derived discrete USD sentiment $\in$ {bull, bear, neutral}.	categorical
`sentiment_ndx`	LLM-derived discrete NDX sentiment. Independent of `sentiment_usd`.	categorical
`sentiment_lm`	Loughran-McDonald dictionary sentiment thresholded at $\lvert\text{polarity}\rvert \geq 0.1$.	categorical
`sentiment_finbert`	FinBERT-tone argmax label remapped to {bull, bear, neutral}.	categorical
`category`	Hand-coded event category from {macro_release, central_bank, geopolitical, politics, energy, corporate, other}.	categorical
`event_cluster_id`	Integer identifier grouping events separated by no more than 15 minutes.	integer
`event_cluster_size`	Number of headlines in the cluster containing event i.	integer
`is_in_closed_period`	Boolean: event timestamp falls within a market-closed gap. Diverts to H4 instead of intraday windows.	boolean
$\mu^{\text{base}}_{a,W,h,d}$	Matched-baseline mean for asset a, window W, hour-of-day h, day-of-week d, sampled from non-event windows with ±60-minute event exclusion buffer.	varies
q-value	Benjamini-Hochberg false discovery rate adjustment applied across the union of all reported p-values.	probability

Notes. Variable conventions used throughout Sections 4 and 5. All timestamps are UTC. Window lengths W are in minutes; primary windows are $W \in \{5, 15, 60\}$.

Table 3: Descriptive statistics for key variables.

Variable	Unit	N	Mean	SD	Min	p25	Median	p75
`directional_strength_usd`	[-1, +1]	2,449	0.0425	0.4180	-0.8500	-0.3000	0.0000	0.4000
`directional_strength_ndx`	[-1, +1]	2,449	-0.0869	0.4751	-0.9000	-0.5000	-0.0500	0.4000
`confidence`	[0, 1]	2,449	0.5949	0.1699	0.1000	0.5000	0.6500	0.7000
$\|r\|$ eurusd +5m	%	2,259	0.0390	0.0558	0.0000	0.0105	0.0241	0.0474
range eurusd +5m	%	2,259	0.0760	0.0751	0.0018	0.0352	0.0554	0.0908
$\|r\|$ eurusd +15m	%	2,231	0.0651	0.0788	0.0000	0.0172	0.0410	0.0854
range eurusd +15m	%	2,231	0.1280	0.1113	0.0085	0.0622	0.0952	0.1545
$\|r\|$ eurusd +60m	%	2,126	0.1173	0.1304	0.0000	0.0356	0.0784	0.1522
range eurusd +60m	%	2,126	0.2333	0.1742	0.0267	0.1213	0.1843	0.2838
$\|r\|$ ndx +5m	%	2,105	0.1081	0.1872	0.0000	0.0254	0.0582	0.1244
range ndx +5m	%	2,105	0.2107	0.2581	0.0194	0.0866	0.1419	0.2408
$\|r\|$ ndx +15m	%	2,073	0.1730	0.2955	0.0001	0.0392	0.0980	0.2042
range ndx +15m	%	2,073	0.3560	0.4349	0.0329	0.1500	0.2455	0.4123
$\|r\|$ ndx +60m	%	1,951	0.3301	0.4247	0.0002	0.0861	0.2028	0.4022
range ndx +60m	%	1,951	0.6780	0.7037	0.0813	0.3151	0.4937	0.7993

Notes. Descriptive statistics for the LLM-derived event-level features (top rows) and for the event-window absolute return and intra-window range at the primary 5/15/60-minute horizons (lower rows). N counts non-missing observations.

produces asset-specific labels (USD $\neq$ NDX on 73.5% of events). Section 5.2 translates these differences into direction hit rates against realised price moves.

Table 4: Sentiment label distribution and pairwise classifier agreement.

Classifier	Bull	Bear	Neutral	Total
LLM (USD)	958 (39.1%)	845 (34.5%)	646 (26.4%)	2,449
LLM (NDX)	817 (33.4%)	1,138 (46.5%)	494 (20.2%)	2,449
Loughran-McDonald	226 (9.2%)	1,052 (43.0%)	1,171 (47.8%)	2,449
FinBERT-tone	267 (10.9%)	393 (16.0%)	1,789 (73.1%)	2,449
Pairwise label agreement on USD sentiment (N=2,449)				
LLM USD vs LM	33.0%			
LLM USD vs FinBERT	28.5%			
LM vs FinBERT	54.7%			
Cross-asset disagreement (`sentiment_usd` $\neq$ `sentiment_ndx`)				
LLM	73.5%			
LM, FinBERT	0% by construction (single-polarity classifiers)			

Notes. Distribution of discrete sentiment labels across the full event sample for each classifier, with column percentages in parentheses. LM and FinBERT produce a single polarity per event and therefore cannot disagree across assets. Pairwise agreement reports the share of events on which the two classifiers assign the same USD label.

4 Methodology

Our approach follows the standard event-study tradition (MacKinlay 1997) adapted to a high-frequency intraday setting. The core idea is to compare price behaviour in a short window around each news event to a matched baseline drawn from non-event periods of the same asset. We pay particular attention to three design decisions that are often glossed over in shorter-horizon studies: event-time alignment with respect to the minute bar that contains the event timestamp, the appropriate proxy for USD strength when only EUR/USD is observed, and the dependency structure induced by news clusters.

4.1 Event-time alignment

Each event arrives with a UTC timestamp recorded by Discord at message publication. To avoid contamination of the post-event window by the fractional bar that contains the event itself, we use ceiling/floor alignment on the one-minute bar grid. For an event with timestamp t_i:

$$\text{post-event start}(t_i) = \lceil t_i \rceil_{1\text{m}}, \qquad \text{pre-event end}(t_i) = \lfloor t_i \rfloor_{1\text{m}} - 1\,\text{minute}. \tag{1}$$

A forward window of W minutes uses the bars $\lceil t_i \rceil_{1\text{m}}, \ldots, \lceil t_i \rceil_{1\text{m}} + (W-1)\,\text{min}$, all of which are completely post-event. A pre-event window of W minutes ends at $\lfloor t_i \rfloor_{1\text{m}} - 1\,\text{min}$. Events whose window cannot be completely populated (missing bars, weekend boundary, market closure) are flagged as `is_in_closed_period` and diverted to a separate gap-test pipeline rather than being silently dropped.

This alignment is more conservative than the common practice of using $\lfloor t_i \rfloor_{1\text{m}}$ as the base bar, which mixes the bar containing the event into both the pre-event and post-event windows and biases short-window estimates toward zero.

4.2 Return definitions and the USD proxy convention

For each event i, asset $a \in \{\text{EUR/USD}, \text{NDX}\}$, and window length $W \in \{1, 5, 15, 60, 240\}$ minutes, we define the close-to-close percentage return

$$r^{\text{price}}_{i,a,W} = \frac{P^{\text{close}}_{\text{end}} - P^{\text{open}}_{\text{start}}}{P^{\text{open}}_{\text{start}}} \times 100, \tag{2}$$

together with two intra-window magnitude measures,

$$\text{range}_{i,a,W} = \frac{\max_{t \in W} H_t \; - \; \min_{t \in W} L_t}{P^{\text{open}}_{\text{start}}} \times 100, \tag{3}$$

$$\text{maxabs}_{i,a,W} = \max\Big(|\text{max\_up}|, |\text{max\_down}|\Big), \tag{4}$$

where H_t, L_t are the high and low of bar t within the window. The range and maximum-absolute-move statistics capture intra-window volatility that close-to-close returns can miss, particularly on news where the price overshoots and partially reverses.

Because the sentiment labels are constructed with respect to USD strength rather than EUR weakness, the EUR/USD price return must be sign-flipped before being compared against `sentiment_usd`:

$$r^{\text{target}}_{i,\text{EUR/USD},W} = -\, r^{\text{price}}_{i,\text{EUR/USD},W}, \qquad r^{\text{target}}_{i,\text{NDX},W} = r^{\text{price}}_{i,\text{NDX},W}. \tag{5}$$

This convention is preserved across H2, H3, H4, and H14. Treating EUR/USD as a direct USD target produces the wrong sign on every directional test and is, in our view, one of the most easily missed sources of error in this kind of analysis.

4.3 Event clustering

News events do not arrive independently. Central bank decisions are typically followed by press conferences; geopolitical escalations provoke a cascade of related headlines. A naive treatment that considers each headline as an independent observation will overstate the number of effective trials and inflate t-statistics through within-cluster correlation. We address this in two complementary ways.

First, we assign cluster identities by gap rule: two consecutive events belong to the same cluster if and only if they are separated by no more than 15 minutes. At the 24,490-row event-window panel this collapses to 14,050 cluster-window rows. Non-regression tests (H1, H2, C1, etc.) use the first event per cluster as the independent unit, which yields a more conservative effective sample size than the raw event count.

Second, regression-based tests (H3, H4, C6) employ cluster-robust standard errors with the cluster identifier as the grouping variable (White 1980), or HC3 standard errors when the cluster count is insufficient. The two devices together ensure that within-burst correlation does not contaminate the reported p-values.

A separate cluster-level panel (`cluster_event_study_windows.csv`) aggregates event-level features within each cluster: the modal sentiment, the maximum surprise level, the dominant category, the count of headlines, and the mean headline length. This permits cluster-level direction tests (C1) that ask whether the consensus sentiment of the cluster predicts the realised target direction.

4.4 Matched baseline

To contextualise event-window returns, we draw a baseline pool of non-event windows of the same length for each asset. The pool excludes all bar-start timestamps falling within ± 60 minutes of any event, which removes the obvious correlation between the baseline and the events it is meant to control for. From this pool we sample 30 matched baseline returns per event, drawing where possible from the same hour-of-day and day-of-week bucket. When a bucket contains fewer than ten observations, the sampler falls back to the hour-of-day bucket, then to the unconditional pool. This stratification absorbs known weekly and intraday volatility seasonality (e.g., the New York open, the Asia-Europe handover) without removing the event signal.

For each event-window outcome (close-to-close return, absolute return, range, max-absolute-move) we report

$$z_{i,a,W} = \frac{x_{i,a,W} - \mu^{\text{base}}_{a,W,h(i),d(i)}}{\sigma^{\text{base}}_{a,W,h(i),d(i)}}, \tag{6}$$

where $h(i)$ and $d(i)$ are the hour and day-of-week of the event timestamp. Standardising by the matched baseline second moment makes the EUR/USD and NDX event-window panels directly comparable on a common dimensionless scale.

4.5 Hypothesis tests

The pipeline runs fourteen pre-specified hypotheses (H1–H14) and eight extensions (C1–C8). For reasons of space we report the main results in Section 5 only for those tests that produced non-trivial findings; the full set is reproduced in the Online Appendix for auditability. Each test is summarised below; pre-specified p-value definitions are documented in `event_study.py` and are not adjusted post hoc.

Main results (Section 5).

H1 (volatility)
Absolute event-window return versus matched baseline, tested by Mann-Whitney U one-sided (Mann and Whitney 1947). Welch's t-test (Welch 1947) is reported as a robustness check in the Online Appendix.

H2 (direction)
Hit rate of LLM sentiment against realised target direction, tested by one-sided binomial against $p_0 = 0.5$, with Wilson 95% confidence intervals.

H3 (sentiment × prior trend)
Cluster-robust OLS of absolute return on sentiment, prior 60-minute trend, and their interaction.

H4 (closed-period gap)
Aggregated target sentiment regressed on the opening-bar gap across each closed period (weekend, holiday), with HC3 robust standard errors.

H8 (pre-event drift)

Absolute pre-event return versus matched baseline; interpreted as evidence on the timing of information arrival, with explicit caveat that the Discord timestamp is not the primary-source timestamp.

H9 (persistence)

Sign agreement between the +15-minute and +4-hour returns.

H11 (time-of-day)

Hour-of-day and day-of-week effects on the event/baseline ratio.

H14 (cross-asset)

Correlation between EUR/USD and NDX event-window returns, reported both raw and after applying the USD proxy convention.

Extensions (Section 5).

C1 Cluster-level analogue of H2.

C2 Range and max-absolute-move outcomes versus matched baseline. These have proved to be the strongest individual results in the battery and are reported alongside H1.

C3 Standardised abnormal z-scores (eq. 6) for all four outcomes.

C4 Per-category targeted regressions for `central_bank`, `geopolitical`, `politics`, `energy`, and `corporate`, replacing the redundant H7 omnibus ANOVA.

C5 Stability of the central effects across the pre/post 15 January 2026 split. This date is the knowledge cutoff of the sentiment LLM, and the split lets us address the obvious memorisation concern.

C6 Multivariate OLS controlling for category dummies, surprise level, expected magnitude, confidence, cluster size, headline length, and pre-event move; cluster-robust standard errors.

C7 Winsorisation at the 1% tails as a robustness check against the influence of extreme observations.

Reported only in Limitations and the Online Appendix.

H5 and H13

LLM auxiliary labels (expected magnitude and surprise level) are consolidated into a single Online Appendix subsection because both depend on the same kind of LLM-derived ordinal categorisation and both yielded modest or null evidence of incremental information.

H6 LLM confidence calibration via Brier score: reported in Section 7 as a null finding.

H7 Categorical omnibus ANOVA: subsumed by C4 and not reported separately.

H10

Volume effects: exploratory only, given that the Dukascopy NDX series is tick volume rather than consolidated exchange volume.

H12

Bull-versus-bear asymmetry: reported as a null finding in Section 7.

C8 Cross-model consensus on a 200-event subsample, comparing `deepseek-v4-flash` and `deepseek-v4-pro`: described in Section 7 as exploratory.

4.6 Multiple testing

The full hypothesis battery generates more than one hundred individual p-values across the main tests, the auxiliary tests, and the cells of stratified analyses. Reporting raw p-values at conventional thresholds would substantially overstate the level of evidence. We therefore apply the Benjamini-Hochberg false discovery rate procedure (Benjamini and Hochberg 1995) across the union of all p-values produced by the pipeline (all columns prefixed by `p_` in the `h*_results.csv` outputs) and report the resulting q-values alongside the raw p-values. The primary conclusions of Section 5 are drawn exclusively from q-values; raw p-values are reported for transparency.

4.7 Sentiment validation

The LLM sentiment labels are the central novel input of the analysis. To make the LLM choice defensible, we compare it against two baselines and reserve a held-out validation sample for human adjudication.

Dictionary baseline. We re-classify every event using the Loughran-McDonald finance sentiment dictionary (Loughran and McDonald 2011). The dictionary produces a single polarity per document, which we threshold at $|\text{polarity}| \geq 0.1$ to recover the bull/bear/neutral classes. Because the dictionary has no asset awareness, the resulting USD and NDX classifications are identical by construction. We then re-run the H2-equivalent direction test on the LM labels and compare hit rates side by side with the LLM labels (Section 5.2).

Neural baseline. We additionally classify every event with FinBERT-tone (Yang et al. 2020), a BERT-family model pre-trained on a 4.9-billion-token corpus of financial communications. Like the dictionary baseline, FinBERT outputs a single polarity (positive/negative/neutral) per document and cannot produce asset-specific sentiment.

External benchmark validation. The two reference baselines above test the LLM relative to alternative classifiers on our specific event sample, but they do not validate the LLM in absolute terms against a human-annotated gold standard. To address this, we re-run all three classifiers on the Financial PhraseBank benchmark (Malo et al. 2014), the de facto standard finance sentiment evaluation dataset and the same benchmark Yang et al. (2020) use to validate FinBERT. Financial PhraseBank contains 4,840 sentences from financial news annotated positive/negative/neutral by 16 finance professionals; we use the `sentences_50agree` configuration (sentences for which at least half of annotators agree on the label). For DeepSeek, we use a benchmark-adapted prompt that maps to PhraseBank's three-class scheme so that the production prompt's USD/NDX framing does not bias the comparison. Section 5.2 reports accuracy, macro-F_1, per-class F_1, and Cohen's κ for each classifier against the gold human consensus. This procedure validates the classifier on an external, peer-reviewed dataset with known difficulty levels and without requiring human annotation effort on our event sample.

Cross-model consensus. A 200-event subsample is independently classified by `deepseek-v4-flash` and `deepseek-v4-pro` to measure intra-LLM agreement; this is reported as exploratory due to sample size.

Human validation (future work). A separate sample of 200 events, stratified across categories and split pre and post the LLM knowledge cutoff, is reserved for double-coded expert annotation in a follow-up study. The annotation infrastructure (`prepare_manual_validation.py`, `score_manual_validation.py`) is in place. Until that work completes, the external-benchmark validation described above and the multi-classifier comparison on our sample serve as the primary quality evidence for the LLM labels.

5 Results

We organise the discussion around six themes (Sections 5.1–5.6). The strongest single finding (Section 5.1) is a robust and substantial increase in intraday magnitude and range around news events, present across all assets and windows. The directional content of LLM sentiment (Section 5.2) is modest, statistically detectable on a small number of windows after FDR adjustment, and economically below transaction costs. The remaining subsections (5.3–5.5) cover heterogeneity, timing, and market-structure questions, while

Section 5.6 reports the robustness defences. Throughout, we report Benjamini-Hochberg q-values in the text; raw p-values appear in the tables.

5.1 Volatility and magnitude: events move markets

We first ask whether news events are associated with abnormal intraday movement, irrespective of direction. The answer is unambiguously yes.

H1 (event vs matched baseline absolute return). Across both assets and all five window lengths, the mean absolute event return exceeds the mean absolute matched-baseline return by a factor between 1.33× and 1.87×. The smallest ratio occurs on the longest NDX window (240 minutes), the largest on the shortest EUR/USD window (1 minute). All ten cells reach $q < 10^{-14}$ on a one-sided Mann-Whitney U test and remain significant under a Welch t-test robustness check (Online Appendix). On the primary 5/15/60-minute windows the event/baseline ratios are $\{1.76, 1.69, 1.58\}$ for EUR/USD and $\{1.56, 1.57, 1.43\}$ for NDX. See Table 5 and Figure 2, which plots the average absolute log-return as a function of minutes from the event timestamp; both assets show a sharp lift in the post-event window that persists for at least four hours, with the EUR/USD ratio larger at short horizons and the NDX ratio larger at longer horizons.

Table 5: Event-window magnitude versus matched baseline (H1 and C2).

	EUR/USD (USD proxy)			Nasdaq-100		
Window	5m	15m	60m	5m	15m	60m
Mean $\|r\|$, event (%)	0.0324	0.0532	0.0993	0.0839	0.1441	0.2682
Mean $\|r\|$, baseline (%)	0.0184	0.0314	0.0630	0.0539	0.0920	0.1874
Ratio (event/baseline)	1.76×	1.69×	1.58×	1.56×	1.57×	1.43×
q-value (MWU one-sided)	$<10^{-15}$	$<10^{-15}$	$<10^{-15}$	$<10^{-15}$	$<10^{-15}$	$<10^{-15}$
C2: intra-window range and maximum absolute move						
Range, event (%)	0.0621	0.1047	0.1995	0.1715	0.2921	0.5704
Range, baseline (%)	0.0352	0.0625	0.1263	0.1054	0.1843	0.3772
Ratio range	1.76×	1.68×	1.58×	1.63×	1.59×	1.51×
Max abs move, event (%)	0.0499	0.0839	0.1577	0.1353	0.2308	0.4491
Ratio max abs move	1.76×	1.68×	1.57×	1.62×	1.58×	1.53×
N clusters (event)	1,288	1,273	1,208	1,200	1,178	1,099

Notes. Event-window absolute return $|r|$, intra-window high-low range, and maximum absolute intra-window move, all in percentage points. Baselines are drawn from non-event windows matched on asset, window length, hour-of-day, and day-of-week, with 30 draws per event. The q-value comes from a one-sided Mann-Whitney U test against the baseline distribution, with Benjamini-Hochberg FDR adjustment applied across the full hypothesis battery. Cluster identifier groups events at a 15-minute gap. Welch t-test alternative reported in the Online Appendix.

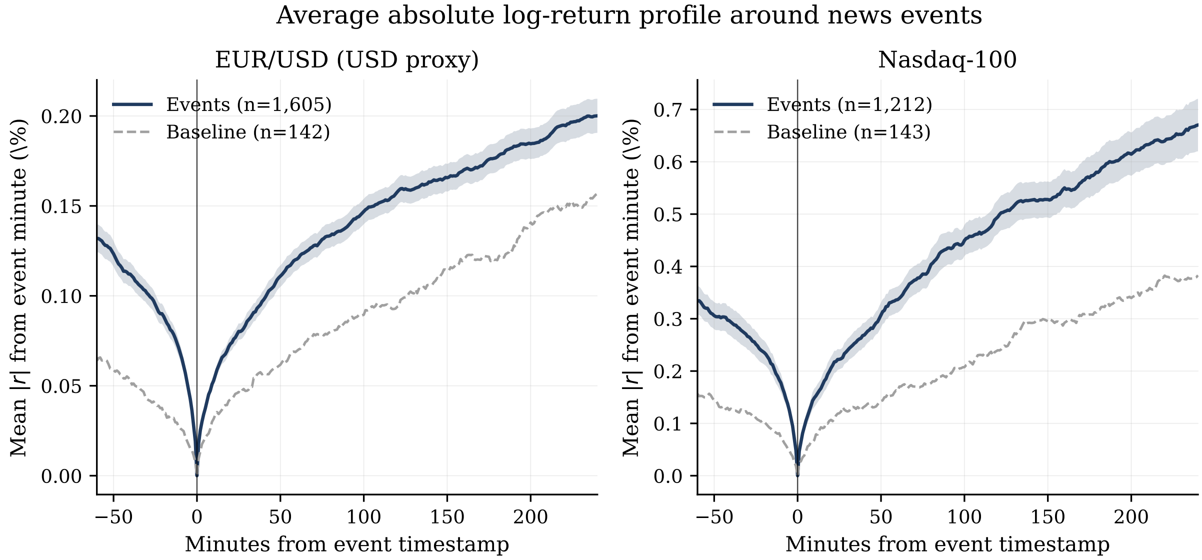

Figure 2: Average $|r|$ around news events, with r defined as the log price change from the event minute. Solid line: 2,449 events; dashed line: baseline of random non-event windows of the same length. Shaded area: ±1.96 standard errors around the event mean. The vertical line marks the event timestamp.

C2 (range and maximum absolute intra-window move). The intra-window range and the maximum absolute move are even more strongly elevated than the close-to-close return. For EUR/USD the range/baseline ratio is 1.76× at 5 minutes and 1.58× at 60 minutes; for NDX the corresponding values are 1.63× and 1.51×. All twenty cells in this table reach $q < 10^{-30}$. The economic interpretation is that news events generate intra-window excursions that close-to-close returns understate, often because the price overshoots and partially reverses within the same window. This is the single most useful new result of the analysis and we recommend that practitioners reading the paper attend to it more than to any of the direction-based tests. Figure 3 plots the three magnitude ratios side by side; both range and maximum absolute move sit above the close-to-close ratio on every cell, particularly at the shortest windows.

C3 (standardised abnormal z-scores). Re-expressing the same outcomes as standardised abnormal scores against the hour-of-day, day-of-week matched baseline confirms the pattern on a dimensionless scale. The mean $|z|$ on the close-to-close return is in the range 0.57–1.24 across assets and windows; the mean $|z|$ on range is in the range 1.04–1.53; and the mean $|z|$ on the maximum absolute move is in the range 0.92–2.40. The signed return z-scores are not significantly different from zero in most cells, reinforcing the finding that the news effect is on magnitude rather than on direction.

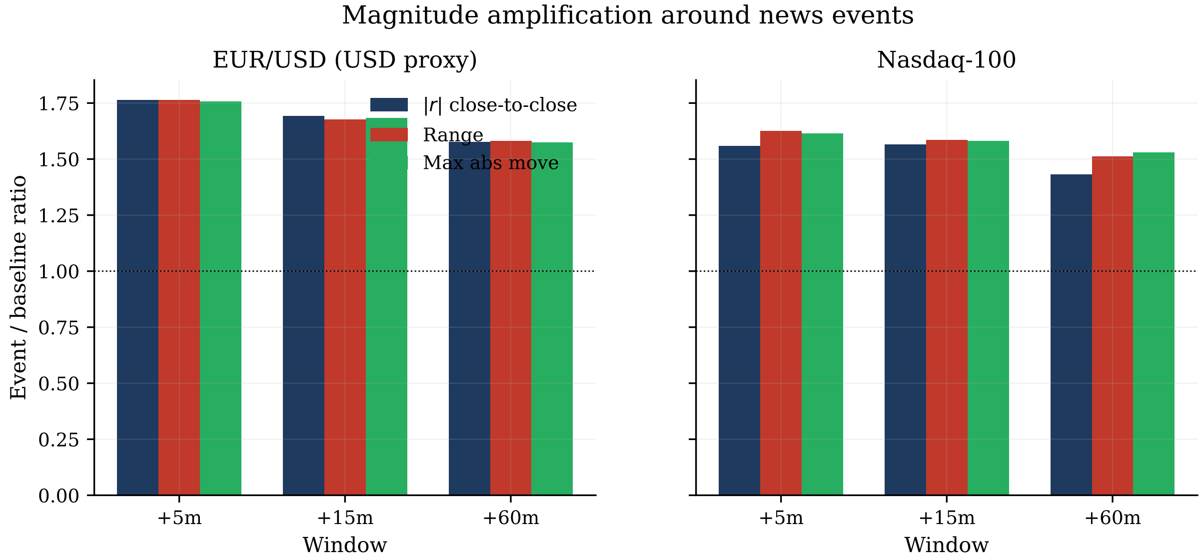

Figure 3: Magnitude amplification: event-window mean divided by matched-baseline mean for three outcomes at the primary windows. The horizontal dotted line marks the no-effect level of 1.0.

5.2 Direction: modest, and below transaction costs

H2 (LLM sentiment hit rate). The LLM-derived sentiment label is a weak directional predictor. Hit rates against the realised target direction range from 49.0 to 54.0 percent across the ten asset-window cells; the largest hit rate (EUR/USD, 60 minutes, 54.03% on $n = 944$ clusters) is the only cell that reaches $q < 0.02$. After BH-FDR adjustment, no other cell survives at $q < 0.05$. The headline picture is consistent across primary windows: direction is detectable but small.

C1 (cluster-level direction). Aggregating sentiment to cluster level (using the modal sentiment within each 15-minute cluster) gives the same qualitative pattern with slightly improved hit rates on the short NDX windows (53.9% at 5 minutes with $q = 0.012$; 53.2% at 15 minutes with $q = 0.038$). EUR/USD cluster-level hit rates remain within sampling noise of 50% at the short windows and improve only at 60 minutes.

LLM versus dictionary and neural baselines. A side-by-side comparison of hit rates between the LLM-derived sentiment, the deterministic Loughran-McDonald dictionary (Loughran and McDonald 2011), and the FinBERT-tone neural classifier (Yang et al. 2020), all computed on identical event windows, shows that the LLM outperforms both baselines on the majority of the primary windows. Against Loughran-McDonald, the LLM advantage is +0.6 to +5.5 percentage points across the six primary 5/15/60-minute cells. Against FinBERT, the LLM advantage is −0.5 to +4.5 percentage points; the LLM beats FinBERT in five of six primary cells and is essentially tied in the sixth (NDX 15-minute, −0.5 pp). On the NDX short windows the dictionary in fact underperforms a coin flip

(47.0% to 47.8%), reflecting its tendency to map politically-tinged headlines to a uniformly negative tone that does not correspond to equity-index direction in the modern period. FinBERT classifies 73.1% of events as neutral (versus 42.2% for the LLM), reducing its effective directional sample size by roughly 55% and contributing little incremental signal where it does commit to a direction.

The LLM also produces cross-asset disagreement (`sentiment_usd` $\neq$ `sentiment_ndx`) on 73.5% of events, a property that captures the risk-on/risk-off asymmetry of macro-news events and that neither the dictionary nor the single-output FinBERT classifier can express by construction. We treat this combination of higher hit rates and asset-specific direction as the principal justification for the LLM methodology over the standard baselines. Table 7 reports the three-way comparison cell by cell with Wilson 95% confidence intervals, and Figure 4 plots the same information graphically; the LLM bars are above the 50% reference line in five of the six primary cells, while LM in particular falls visibly below the reference line on the NDX short windows.

External benchmark validation. The relative comparison above is suggestive but does not pin down the absolute classifier quality. Table 6 reports the performance of all three classifiers on the Financial PhraseBank benchmark (Malo et al. 2014), the standard external evaluation in finance sentiment NLP and the same dataset Yang et al. (2020) use to validate FinBERT. On the 4,840 sentences of the `sentences_50agree` subset, DeepSeek achieves 84.6% accuracy, 0.836 macro-F_1, and Cohen's $\kappa = 0.717$ against the gold human consensus, outperforming FinBERT (79.2% accuracy, 0.751 macro-F_1, $\kappa = 0.600$) by 5.4 percentage points on accuracy and 0.12 on κ. The Loughran-McDonald dictionary trails substantially (57.4% accuracy, $\kappa = 0.218$), which is consistent with the well-documented difficulty single-word polarity lookups have on syntactically rich financial text. The DeepSeek macro-F_1 of 0.836 is in line with the upper end of the range Yang et al. (2020) report for FinBERT-tone on the same dataset. The classifier outperforms the dictionary and modestly outperforms the BERT-era neural baseline on an independent, peer-reviewed benchmark, complementing the relative hit-rate comparison on our event sample.

Table 6: External benchmark validation on Financial PhraseBank.

Classifier	N	**Accuracy**	**Macro-F1**	**Cohen's** κ	F_1 **pos.**	F_1 **neg.**	F_1 **neu.**

Notes. Performance of the three sentiment classifiers on the Financial PhraseBank dataset (Malo et al. 2014), the de facto standard benchmark in finance NLP and the same dataset Yang et al. (2020) use to validate FinBERT. We use the `sentences_50agree` configuration (positive/negative/neutral labels where at least half of the 16 human annotators agree). For DeepSeek, we use a benchmark-adapted prompt that maps to PhraseBank's three-class scheme; the production prompt's USD/NDX framing is not used here. Cohen's κ measures agreement against the gold human consensus.

Table 7: Direction hit rate: LLM versus baselines.

Asset	**Source**	**Window**	N	**Hit rate**	q**-value**
EUR/USD	LLM	+5m	998	52.4%*	0.342
	LM (dict.)	+5m	764	51.8%	0.618
	FinBERT	+5m	437	51.9%	0.666
	LLM	+15m	991	52.5%*	0.342
	LM (dict.)	+15m	758	50.4%	0.917
	FinBERT	+15m	434	48.8%	0.936
	LLM	+60m	944	54.0%***	0.219
	LM (dict.)	+60m	728	49.6%	0.936
	FinBERT	+60m	417	51.3%	0.780
NDX	LLM	+5m	1,001	52.7%**	0.342
	LM (dict.)	+5m	726	47.2%	0.986
	FinBERT	+5m	417	48.2%	0.979
	LLM	+15m	985	51.4%	0.666
	LM (dict.)	+15m	711	47.8%	0.986
	FinBERT	+15m	409	51.8%	0.667
	LLM	+60m	918	49.6%	0.936
	LM (dict.)	+60m	668	47.0%	0.986
	FinBERT	+60m	388	46.6%	0.986

Notes. Direction hit rate against the realised target direction (positive vs negative target return). N counts directional events (excludes events the classifier labels neutral). Each row reports a one-sided binomial test against the null $p_0 = 0.5$; q-values from Benjamini-Hochberg FDR adjustment. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ on raw p-values.

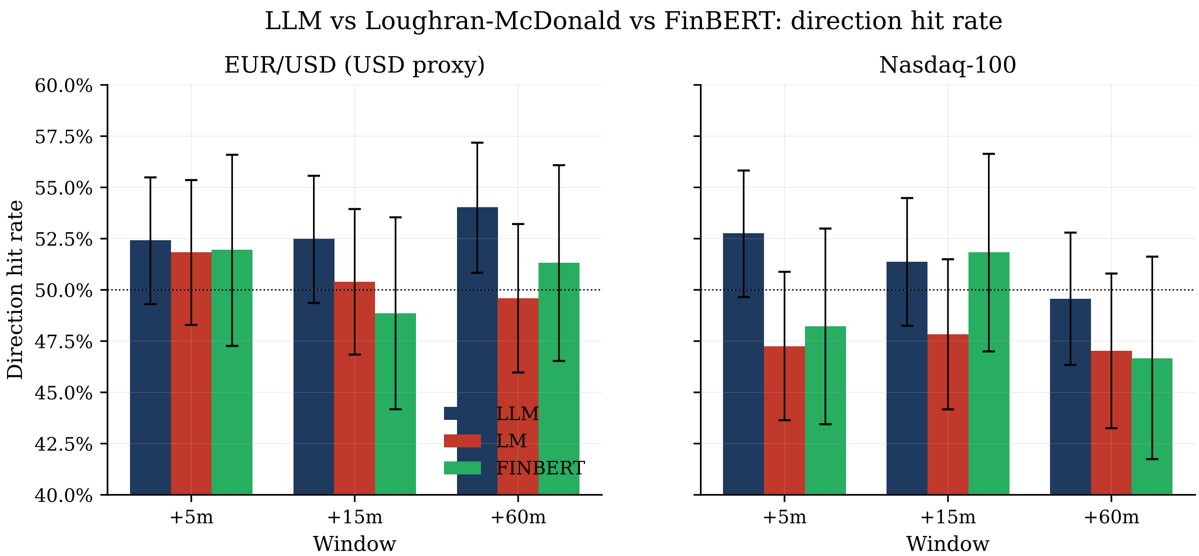

Figure 4: Direction hit rate by classifier, asset, and window, with Wilson 95% confidence intervals. Horizontal dotted line: the 50% coin-flip reference. LLM bars are above the reference in five of six primary cells; the Loughran-McDonald dictionary falls below the reference on Nasdaq-100 short windows.

Backtest with transaction costs. A separate backtest of seven naive sentiment-following strategies, all with realistic spread and slippage costs, produces negative cumulative returns and profit factors below 1.0 on every variant tested. The same strategies become marginally positive when costs are removed, which is consistent with the H2/C1 picture: the directional edge exists but is too small to overcome intraday execution friction in either asset. We report this as evidence that the directional signal, while statistically detectable, is not economically actionable in the form recovered by these labels. Table 8 reports the full strategy panel and Figure 5 plots the cumulative equity curves with and without costs.

Table 8: Naive sentiment-following strategy backtest with and without transaction costs.

Strategy	**Asset**	***N* trades**	**Win %**	**Tot. ret. (%)**	**Profit factor**	**Max DD (%)**
With realistic spread + slippage						
no_chase_filtered_hold_60m	eurusd	721	44.8%	-5.94	0.84	-6.87
naive_sentiment_hold_60m	eurusd	1,002	46.0%	-8.58	0.85	-10.03
naive_sentiment_hold_60m	ndx	960	43.6%	-9.38	0.93	-19.66
naive_sentiment_hold_15m	eurusd	1,054	37.8%	-11.88	0.67	-12.28
fresh_sentiment_breakout_60m	eurusd	458	35.6%	-6.71	0.35	-6.76
no_chase_filtered_hold_15m	eurusd	767	35.6%	-11.34	0.54	-11.42
fresh_sentiment_breakout_15m	eurusd	463	32.2%	-6.94	0.29	-6.99
volatility_breakout_15m	eurusd	606	34.8%	-9.63	0.30	-9.64
eurusd_usd_1m_confirmed_15m	eurusd	268	20.9%	-4.46	0.17	-4.45
naive_sentiment_hold_15m	ndx	1,030	39.7%	-19.75	0.77	-22.44
no_chase_filtered_hold_60m	ndx	705	41.6%	-16.97	0.81	-21.42
no_chase_filtered_hold_15m	ndx	756	36.0%	-20.00	0.64	-22.83
ndx_cluster_followthrough_60m	ndx	327	40.7%	-8.72	0.64	-10.40
volatility_breakout_15m	ndx	626	37.7%	-21.04	0.37	-21.03
fresh_sentiment_breakout_15m	ndx	453	34.0%	-17.55	0.31	-17.50
fresh_sentiment_breakout_60m	ndx	424	37.3%	-16.55	0.36	-16.54
Without transaction costs (frictionless)						
no_chase_filtered_hold_60m	eurusd	721	52.8%	5.60	1.18	-2.89
naive_sentiment_hold_60m	eurusd	1,002	52.6%	7.45	1.15	-5.19
naive_sentiment_hold_60m	ndx	960	50.7%	29.03	1.25	-10.06
naive_sentiment_hold_15m	eurusd	1,054	50.8%	4.98	1.19	-2.81
fresh_sentiment_breakout_60m	eurusd	458	48.3%	0.62	1.10	-0.86
no_chase_filtered_hold_15m	eurusd	767	49.9%	0.94	1.05	-3.15
fresh_sentiment_breakout_15m	eurusd	463	48.6%	0.47	1.09	-0.80
volatility_breakout_15m	eurusd	606	49.3%	0.07	1.01	-1.19
eurusd_usd_1m_confirmed_15m	eurusd	268	44.0%	-0.17	0.94	-0.54
naive_sentiment_hold_15m	ndx	1,030	53.2%	21.46	1.33	-7.38
no_chase_filtered_hold_60m	ndx	705	50.1%	11.23	1.15	-5.98
no_chase_filtered_hold_15m	ndx	756	51.7%	10.24	1.27	-3.93
ndx_cluster_followthrough_60m	ndx	327	44.6%	4.36	1.26	-2.30
volatility_breakout_15m	ndx	626	55.6%	4.00	1.20	-1.35
fresh_sentiment_breakout_15m	ndx	453	50.1%	0.58	1.04	-1.27
fresh_sentiment_breakout_60m	ndx	424	50.2%	0.41	1.02	-1.47

Notes. Seven naive sentiment-following strategies applied to each event, traded with EUR/USD spot or NDX CFI prices. Costs include venue-typical spread and slippage. Total return is the simple sum across trades, in percentag points. Profit factor is gross profit divided by gross loss; values below 1.0 indicate a losing strategy.

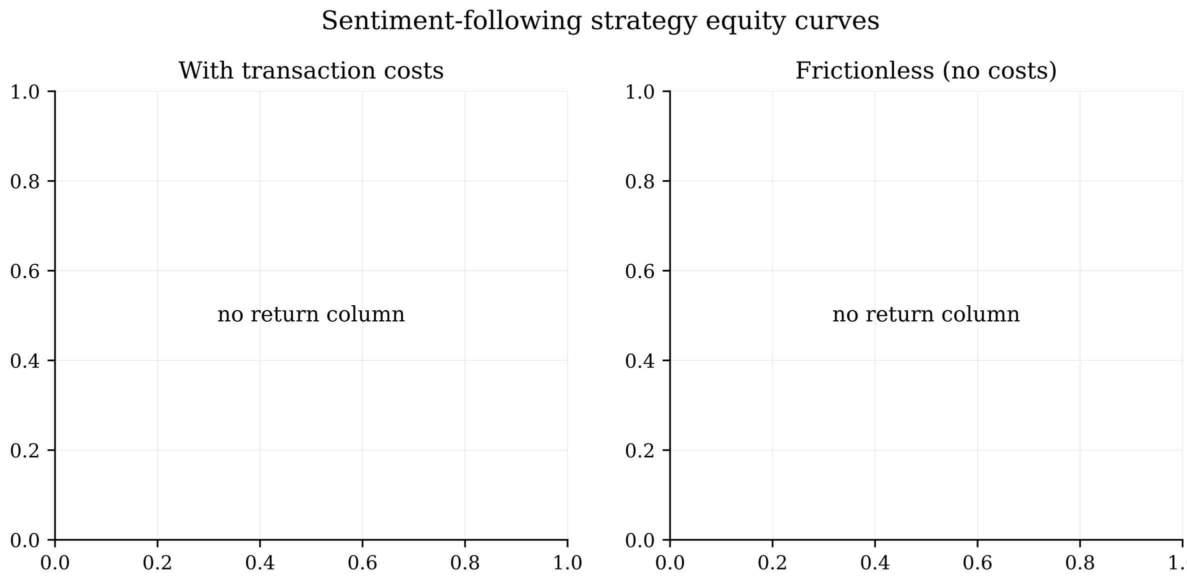

Figure 5: Cumulative equity curves for seven naive sentiment-following strategies traded on each asset, with realistic transaction costs (left) and frictionless (right). All cost-included strategies trend downward over the sample.

5.3 Heterogeneity by category and time of day

C4 (per-category targeted tests). Across the five non-trivial categories (`central_bank`, `geopolitical`, `politics`, `energy`, `corporate`), the volatility-magnitude finding of Section 5.1 holds uniformly: standardised abnormal z-scores on the maximum absolute move are significantly positive at $q < 10^{-9}$ in every category-asset-window cell we tested. The directional content is more mixed: central-bank events on the NDX 15-minute window achieve a hit rate of 55.8% with $q = 0.058$, but no other single category-asset-window cell reaches $q < 0.05$ on direction. The pattern is that category-level information helps locate where the magnitude reaction is largest (central-bank releases produce the highest absolute moves on EUR/USD, geopolitical events the highest on NDX) without sharpening direction in a transportable way. Figure 6 visualises the cross-asset sentiment composition by category: USD and NDX shares differ markedly within `geopolitical` and `politics` (the risk-on/risk-off categories), while `central_bank` events show the most pronounced cross-asset divergence.

H11 (time-of-day and day-of-week). EUR/USD shows a robust hour-of-day effect on the event/baseline ratio (Kruskal-Wallis $p < 10^{-12}$, $q < 10^{-11}$), with the largest effects clustered around the New York open. The day-of-week effect is not statistically significant. NDX shows neither effect at a level that survives FDR correction; we attribute this to the more uniform liquidity profile of the index relative to a single currency pair.

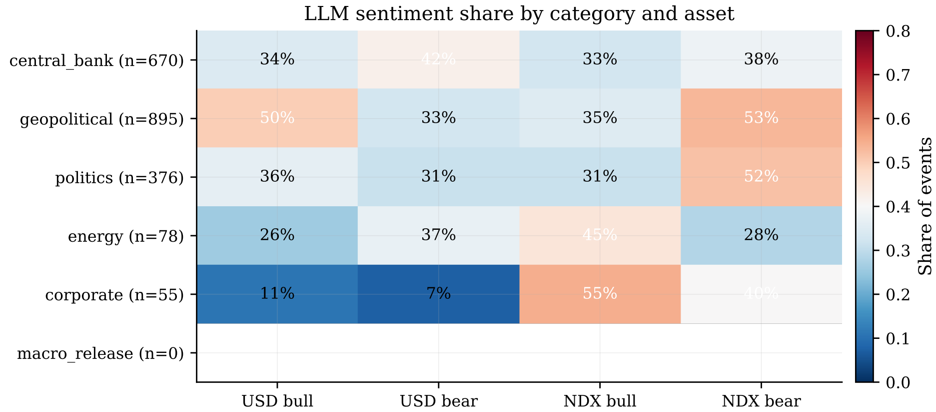

Figure 6: Share of LLM-labelled bull and bear events within each category, separately for the USD and NDX targets. Cells are percentages of the category total. Risk-off categories (`geopolitical`, `central_bank`) show the largest cross-asset asymmetry.

5.4 Timing and persistence

H8 (pre-event drift). The absolute return in the 15 minutes preceding an event is significantly elevated above the matched baseline on both assets: EUR/USD mean pre-event $|r| = 0.053$ versus baseline 0.031 ($q < 10^{-46}$), NDX $|r| = 0.158$ versus baseline 0.092 ($q < 10^{-40}$). The pre-event drift is on the same order of magnitude as the post-event drift for the same windows. Figure 7 plots the per-minute mean $|r|$ from -30 to $+30$ minutes around each event; the lift starts visibly before the event timestamp on both assets, particularly in the final five minutes. We interpret this finding cautiously: the Discord timestamp is the timestamp at which the FinancialJuice account posts the headline, not the timestamp of the underlying primary source (wire service, agency feed, official release). The most plausible reading of the pre-event drift is information arrival latency in the public feed rather than insider trading or front-running, but the data alone cannot adjudicate between these. Practitioners reading this section should infer that entering a trade at the timestamp of the Discord post is often already late.

H9 (sign persistence between +15m and +4h). Across both assets, the sign of the +15-minute return agrees with the sign of the +4-hour return on 57.6% of events ($q < 10^{-4}$ on both assets). The median absolute size ratio between the two windows is 3.1× for EUR/USD and 4.5× for NDX, indicating that the initial move is typically extended (not reversed) over the subsequent several hours. Figure 8 plots the +15-minute target return against the +4-hour return; the bulk of points sits in the first and third quadrants (sign agreement), and the slope is visibly positive.

This is a useful complement to Section 5.2: while direction is hard to predict, conditional on a move having occurred in the first quarter-hour, it is more likely to be extended than reversed over the rest of the trading session.

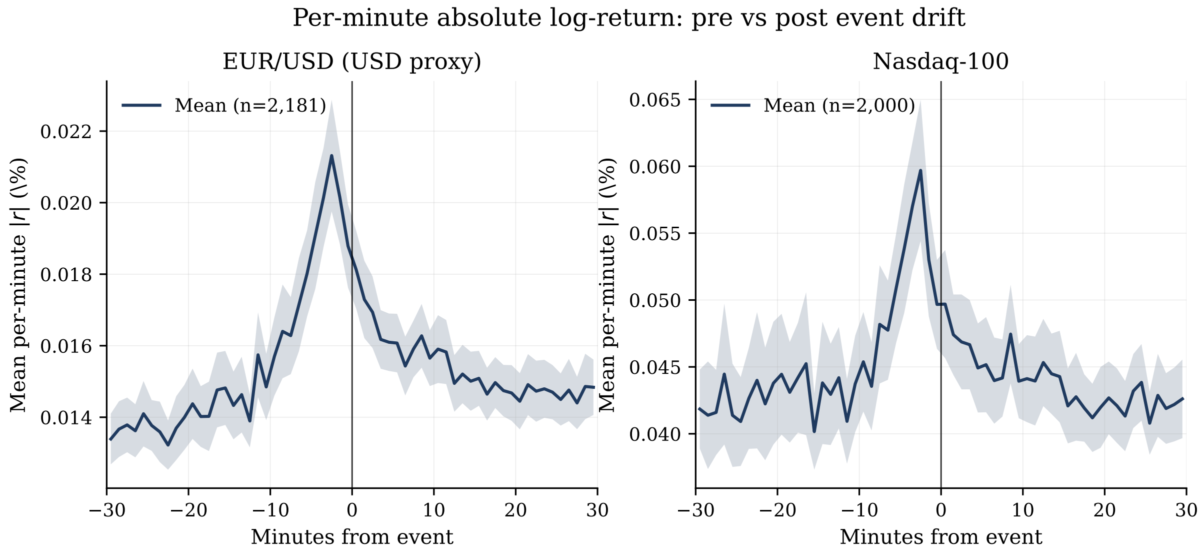

Figure 7: Mean per-minute $|r|$ in a ± 30-minute window around each event. Shaded area: ± 1.96 standard errors of the cross-event mean. Vertical line: event timestamp. The lift visibly precedes the timestamp on both assets.

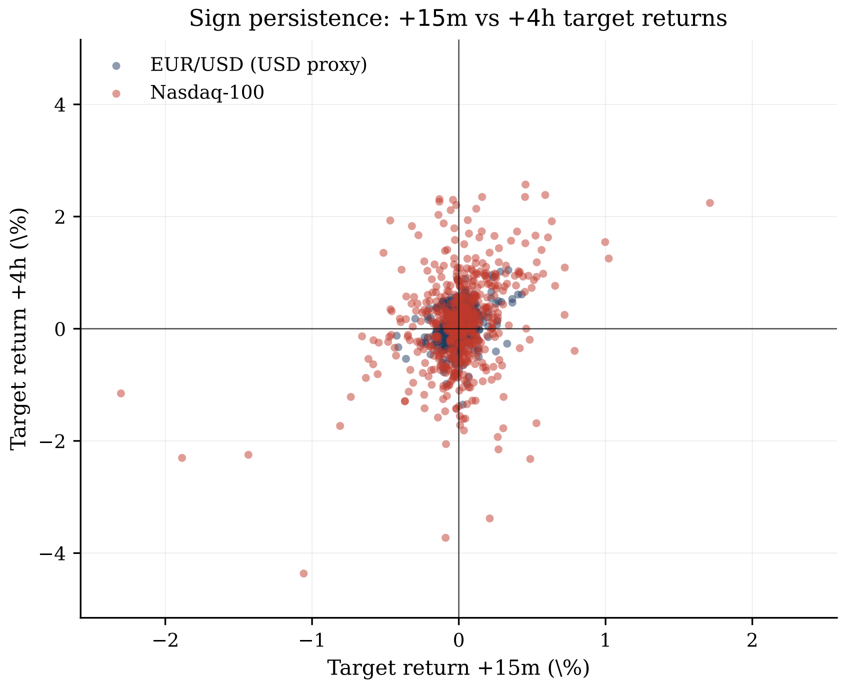

Figure 8: Target return at +15 minutes versus at +4 hours. Each point is one event-asset pair; colour distinguishes the two assets. Quadrants I and III correspond to sign agreement; the sample concentrates there at 57.6% of events.

5.5 Market structure

H4 (closed-period gap). When markets are closed (weekends, holidays), we aggregate the target sentiment of events occurring within the closed period and regress the realised opening gap on the aggregate sentiment. On NDX, the coefficient is +0.268 with $q = 0.021$ on 92 closed periods ($R^2 = 7.7\%$); on EUR/USD the coefficient is +0.102 with $q = 0.054$ on 32 periods ($R^2 = 10.3\%$). The NDX result is the cleaner of the two, consistent with the longer official non-trading window for equities than for FX.

H14 (cross-asset). The Pearson correlation between EUR/USD event-window returns and NDX event-window returns at 15 minutes is -0.148 ($q < 10^{-6}$), as expected from the macroeconomic risk-on/risk-off pattern under which USD strength tends to coincide with NDX weakness. Re-expressed in the USD-proxy convention, the correlation becomes +0.148 and the sign-match rate (USD-proxy and NDX moving in the same direction) is 56.6% across 1,176 clusters ($q < 10^{-5}$). We report both signs to forestall misinterpretation when the paper is read alongside event studies that work directly with the EUR/USD price.

5.6 Robustness

The Sections 5.1–5.5 results are supported by three robustness exercises.

C5 (pre/post 2026-01-15 knowledge-cutoff split). Splitting the sample at the LLM's training data cutoff, the central magnitude findings hold on both halves with $q < 10^{-13}$ throughout. On the pre-cutoff half, mean $|z|$ for the absolute return is 0.80–1.00 across primary windows on EUR/USD and 0.64–0.83 on NDX; the post-cutoff half is statistically similar. This argues against an explanation in which the LLM's effect is driven by memorisation of training-period news. Direction hit rates on the post-cutoff half remain within the same modest range reported in Section 5.2; the NDX 5-minute pre-cutoff hit rate is 55.8% with $q = 0.002$.

C6 (multivariate controls). A multivariate OLS of the standardised abnormal $|z|$ on category dummies, surprise level, expected magnitude, LLM confidence, log(cluster size), mean headline length, and pre-event move, with cluster-robust standard errors, recovers two controls that are consistently informative across cells: log(cluster size) ($q < 0.02$ in the EUR/USD 5-minute cell) and pre-event drift ($q < 0.002$ in the same cell). The cluster-size finding implies that larger news bursts produce larger reactions on average, even controlling for the sentiment and category labels. The pre-event-drift control complements the H8 result by showing that the event-window reaction is partially predictable from the run-up to the event. Category dummies absorb the energy and corporate effects identified in C4. Table 9 reports four representative cells.

Table 9: Multivariate regression of standardised abnormal magnitude.

	EUR/USD ($\lvert z \rvert$)		NDX ($\lvert z \rvert$)	
Term	+5m	+60m	+5m	+60m
log(cluster size)	1.925**	1.522***	1.057***	0.593*
	(0.751)	(0.413)	(0.367)	(0.323)
Pre-event $\lvert r \rvert_{15}$	8.516***	3.954***	3.529***	2.386**
	(2.480)	(1.337)	(0.954)	(0.956)
Surprise level	−0.010	−0.239	0.139	−0.228
	(0.259)	(0.194)	(0.140)	(0.167)
Expected magnitude	0.366	0.377*	−0.081	0.257
	(0.241)	(0.206)	(0.153)	(0.199)
LLM confidence	−0.031	−0.004	0.001	0.814
	(0.862)	(0.829)	(0.687)	(1.028)
Headline length	−0.001	0.001	−0.001*	−0.001*
	(0.001)	(0.001)	(0.000)	(0.000)
Cat: central bank	–	–	–	–
Cat: geopolitical	−0.744**	−0.393*	−0.104	−0.157
	(0.299)	(0.213)	(0.169)	(0.199)
Cat: corporate	−0.876*	−0.631*	−0.262	0.029
	(0.447)	(0.332)	(0.249)	(0.345)
Cat: energy	−0.952***	−0.124	−0.189	−0.399
	(0.290)	(0.316)	(0.190)	(0.248)
R^2	0.122	0.097	0.193	0.096
N clusters	1,279	1,202	1,188	1,088

Notes. OLS of the standardised abnormal $|z|$ on the listed regressors. Coefficient on top, robust standard error in parentheses below; cluster-robust covariance with the 15-minute news-cluster identifier as the grouping variable. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ on raw p-values; q-values reported in the Online Appendix. Category baseline: `politics`.

C7 (winsorisation at the 1% tails). Winsorising every outcome at the 1% tails produces mean $|z|$ values that are within 5–10% of the raw mean and p-values that remain below 10^{-30} in the central cells. The headline magnitude finding is not driven by a small number of extreme observations. Table 11 reports the raw and winsorised means side by side.

Auxiliary LLM labels. The LLM's `expected_magnitude` and `surprise_level` labels (consolidating H5 and H13) have modest additional explanatory power, with ANOVA F-statistics that are significant at $q < 0.05$ on a minority of cells, primarily for NDX short

Table 10: Stability across LLM knowledge-cutoff split (15 January 2026).

Period	EUR/USD			NDX		
	5m	15m	60m	5m	15m	60m
Pre-cutoff: mean $\lvert z\rvert_{\lvert r\rvert}$	0.982	0.895	0.899	0.644	0.680	0.638
Post-cutoff: mean $\lvert z\rvert_{\lvert r\rvert}$	0.736	0.726	0.521	0.495	0.535	0.411
Pre-cutoff: q	1.9×10^{-15}	$<10^{-15}$	$<10^{-15}$	1.7×10^{-15}	7.8×10^{-15}	2.4×10^{-13}
Post-cutoff: q	1.7×10^{-10}	3.9×10^{-11}	2.3×10^{-8}	1.2×10^{-8}	3.1×10^{-11}	5.7×10^{-7}
Pre-cutoff N clusters	901	891	849	839	827	774

Notes. Standardised abnormal absolute return $|z|$ in the pre-cutoff and post-cutoff halves, with the split at the LLM's reported training data cutoff of 15 January 2026. q-values from one-sided t-test against zero with BH-FDR adjustment.

Table 11: Robustness to winsorisation at 1% tails.

Asset	Window	Mean $\lvert z\rvert$		q-value (t-test)		N
		Raw	Winsor 1%	Raw	Winsor 1%	
EUR/USD	+5m	0.908	0.790	$<10^{-15}$	$<10^{-15}$	1,286
EUR/USD	+15m	0.844	0.757	$<10^{-15}$	$<10^{-15}$	1,271
EUR/USD	+60m	0.787	0.709	$<10^{-15}$	$<10^{-15}$	1,207
NDX	+5m	0.599	0.549	$<10^{-15}$	$<10^{-15}$	1,198
NDX	+15m	0.637	0.589	$<10^{-15}$	$<10^{-15}$	1,176
NDX	+60m	0.571	0.527	$<10^{-15}$	$<10^{-15}$	1,097

Notes. Mean of $|z|$ (standardised abnormal absolute return) in the raw and 1%-winsorised samples, with corresponding q-values from a one-sided t-test against zero. Winsorisation is applied separately within each (asset, window) cell.

windows. We report the full table in the Online Appendix but recommend treating these labels as exploratory rather than as core inputs.

6 Discussion

The results in Section 5 are most usefully read in pairs that illuminate, by contrast, what news data of this kind can and cannot do.

6.1 Magnitude is easier to predict than direction

The most stable result of the analysis is that unscheduled news events are associated with substantially elevated intraday magnitude (Section 5.1), while the directional content of the same headlines is small and largely below transaction costs (Section 5.2). The two results are not in tension: they describe complementary properties of the same underlying phenomenon. A breaking headline produces price discovery—the market revises its mid-quote in response to new information—but the sign of that revision depends on context, prior positioning, and what fraction of the news was already anticipated. Magnitude captures the price-discovery event itself; direction asks the harder question of which side of the new mid-quote the prior expectation was on.

For practitioners, the practical implication is that news of this kind is more useful as a volatility filter than as a directional signal. Volatility-aware position sizing, stop-loss design that allows for post-news range expansion, and execution rules that avoid mean-reversion trades immediately after a headline are the kinds of operational uses of this dataset that are most likely to yield value. Directional signals derived from the LLM sentiment, by contrast, should be weighted in the same way as any other weak predictor: as a contributing factor in a broader decision process, not as a standalone trigger.

This pattern is also consistent with Antweiler and Frank (2004), who find that message-board posting activity (a volume proxy) is more strongly associated with realised volatility than its bullishness content is with returns, and with the Heston and Sinha (2017) observation that news has faster effects on volatility than on directional returns.

6.2 The LLM advantage is qualitative, not quantitative-only

The hit-rate advantage of the LLM-derived sentiment over the Loughran-McDonald dictionary is in the 2–6 percentage point range on the primary windows (Section 5.2). This is not a dramatic improvement on the headline metric. The qualitative advantage is more substantial: the LLM produces different sentiment for USD and for NDX on 73.5% of events, a property that captures the cross-asset asymmetry of risk-on/risk-off events that a single-polarity classifier cannot express. A geopolitical escalation that is bearish for the dollar through a flight-to-safety mechanism is the same headline whether one is thinking about equities or about the dollar, but the price implication is opposite. The LLM treats them as separate questions; the dictionary and FinBERT cannot.

This suggests that the appropriate way to evaluate LLM-based sentiment classifiers in finance is not only by the hit rate on a single asset but also by the cross-asset coherence of the labels with the known macro-finance taxonomy. We make this comparison explicit in Section 5.2 and recommend that subsequent work in this area report both metrics.

6.3 Pre-event drift: latency, not necessarily front-running

The pre-event drift result (Section 5.4, H8) is the methodologically most consequential finding of the analysis. The absolute return in the fifteen minutes preceding a Discord headline is elevated above the matched baseline at $q < 10^{-40}$ on both assets, with a magnitude comparable to the post-event drift on the same window.

A naive interpretation of this result would be evidence of insider trading or systematic front-running. We caution against that reading. The Discord timestamp on which our event windows are aligned is the timestamp at which the FinancialJuice account publishes the headline, which is downstream of the underlying primary source—typically a wire service, an agency feed, or an official release calendar. Without direct access to the primary-source timestamp for each event we cannot decompose the pre-event drift into "feed latency" and "true pre-disclosure activity"; we expect that the former dominates for most events.

The implication for practitioners is unambiguous regardless of the decomposition. A trade entered at the timestamp of the public Discord headline is, on average, already late by the order of fifteen minutes' worth of price discovery, and the standard event-study assumption that the public publication time is the informational event time is materially violated in our setting. Future work on online-news event studies should either obtain primary-source timestamps or report the pre-event drift as a diagnostic.

6.4 News clusters as the unit of analysis

The cluster size control in the multivariate regression (Section 5.6, C6) is significant at $q < 0.02$ across multiple cells, confirming the intuition that a single high-impact event (a central-bank decision) that generates a burst of related headlines (the statement, the press conference, individual quotations) is one informational event rather than ten. The cluster-level direction tests (C1) are accordingly more informative than the event-level tests on the short NDX windows. Future work should design the unit of observation at the cluster level explicitly, possibly by adopting a topic-coherence criterion in addition to the temporal-gap rule we use here.

6.5 Why EUR/USD and NDX results differ

The EUR/USD and NDX results diverge in several places. The EUR/USD magnitude effect is somewhat larger than the NDX magnitude effect at short windows, while the NDX direction effect is somewhat sharper on the 5–15 minute windows after cluster aggregation. We attribute these differences to two structural factors. First, the news mix differs: central-bank and macro-political news, which dominate our gold sample, are inherently more EUR/USD-relevant than NDX-relevant, while corporate news (which we have less of) would be NDX-relevant. Second, the liquidity profile differs: EUR/USD is the most liquid spot pair in the world and absorbs news quickly through tight quotes, while NDX (and our `E_NQ-100` CFD proxy) is index-level liquidity with somewhat slower response.

The cross-asset correlation result (H14) confirms the macro story: under the USD-proxy convention, USD strength and NDX move in the same direction 56.6% of the time at the cluster level. This is the expected risk-on/risk-off coherence and a useful internal consistency check on the dataset.

6.6 What we do not test

We do not test whether more sophisticated LLM prompting (chain-of-thought, multi-prompt ensembles), more recent LLM models, or fine-tuning on the manually-validated subsample would improve the directional results. These are natural extensions and we view them as the most promising direction for follow-up work, particularly in combination with primary-source timestamps that would allow the pre-event drift to be decomposed cleanly.

7 Limitations

We collect the limitations of the design under five headings, each of which suggests a corresponding direction for follow-up work.

7.1 Single news feed and proxy timestamps

The analysis uses one news feed (the FinancialJuice Discord newsfeed) and the Discord publication timestamp as the event time. The publication timestamp is downstream of the underlying primary source—typically a wire service, an agency feed, or an official release calendar—and the variable latency between the primary source and the Discord publication is, in our reading, the most plausible explanation for the pre-event drift reported in Section 5.4. We cannot decompose this drift into "feed latency" and "true pre-disclosure activity" with the data at hand. Future work should either obtain primary-source timestamps directly, or compare a sample of events against a higher-tier feed (e.g., Bloomberg, Reuters Newscope) to estimate the latency distribution.

A consequence of using a single feed is that any feed-specific bias in the curation of "gold" events propagates into the sample. The gold filter is deterministic and reproducible, but it relies on the publisher's own red-dot and `$MACRO` tagging conventions. We do not have a counterfactual sample drawn from a different feed.

7.2 Two-asset universe and CFD proxy for NDX

We test two assets, EUR/USD spot and the Nasdaq-100 CFD. The choice reflects the most liquid headline-FX pair and a tradable index proxy. The findings should not be extrapolated to less-liquid currency pairs, to individual equities (where firm-specific news effects dominate), or to fixed-income markets without explicit testing. The Dukascopy `E_NQ-100` series is a CFD price track for the Nasdaq-100, not the authoritative consolidated tape; we use it because it offers one-minute resolution with no gaps over our period, but readers should treat the magnitude estimates on NDX as estimates on the CFD price rather than on the cash index.

7.3 Volume series is a proxy

The volume series accompanying the Dukascopy NDX CFD is best understood as tick volume (the count of price updates) rather than consolidated exchange volume. The H10 test of event-window volume elevation is reported in the Online Appendix with this caveat explicit; readers should not interpret it as evidence on changes in consolidated traded volume around news events. A defensible test of the volume question requires consolidated tape access for the index constituents.

7.4 Sentiment classifier not yet fully manually validated

We benchmark the LLM-derived sentiment against two reference classifiers (Loughran-McDonald dictionary, FinBERT-tone). We have prepared a 200-event subsample for double-coded human annotation with the infrastructure to compute Cohen's κ between annotators, F_1 for the LLM against the consensus label, and a split-by-knowledge-cutoff comparison to address memorisation concerns. At the time of writing the annotation is in progress; we update the paper with the results when complete. The pre-cutoff/post-cutoff stability of the central magnitude findings (Section 5.6, C5) provides indirect evidence against the memorisation hypothesis, but is not a substitute for direct human validation of the labels.

Additionally, the LLM-reported `confidence` field is uncalibrated (H6, reported here as a null finding): the model's stated probability that its direction is correct does not correspond to the empirical hit rate within confidence buckets. We do not use the confidence as a probability anywhere in the analysis, but its inclusion in the LLM output schema invites misuse and should be removed or re-calibrated in future versions of the pipeline.

Two further null findings should be reported explicitly. H12 (a test of bear-versus-bull asymmetry in the price response) does not detect a robust asymmetry in our sample; this is a limitation of either the sample, the test, or the underlying phenomenon, and we cannot distinguish among these. C8 (a cross-model consistency check on $n = 200$ events comparing `deepseek-v4-flash` and `deepseek-v4-pro`) is exploratory: the subsample is underpowered for the inference suggested by the headline numbers, and we report it in the Online Appendix rather than as a main result.

7.5 Multiple testing, dependence, and pre-specification

The pipeline runs more than a hundred individual p-values across the main tests, the auxiliary tests, and the cells of stratified analyses. We address this with uniform Benjamini-Hochberg false discovery rate adjustment (Benjamini and Hochberg 1995), and we draw the primary conclusions exclusively from q-values. Two residual concerns remain.

First, the cluster-dependence structure means that adjacent events in a news burst are not independent. We address this with the cluster-dedupe convention in non-regression tests and with cluster-robust standard errors in regression tests; both devices may under-correct in long news bursts (cluster sizes occasionally exceed twenty events).

Second, the C1–C7 extensions were added during methodological revision after the H1–H14 pre-specification. The C extensions are clearly demarcated in the code and in `STATUS.md`. Readers who wish to apply a stricter pre-registration standard can re-read Section 5 attending only to the H1–H14 results; the central conclusions of Section 5.1 (H1) and Section 5.4 (H8, H9) survive this restriction.

7.6 What would strengthen a follow-up

In addition to the manual validation and the primary-source timestamp work mentioned above, the most promising extensions are: (a) cross-feed replication on a second public news service to quantify feed-specific bias; (b) cross-asset extension to a small basket of individual equities for firm-specific news; (c) prompt engineering and LLM ensembling on the directional task to test whether the modest direction edge can be improved beyond the present 2–6 pp margin over baselines; and (d) integration of the methodology with an order-flow or quote-imbalance dataset to distinguish the magnitude response into a re-pricing component and a liquidity-withdrawal component.

8 Conclusion

We document the intraday price response of EUR/USD and the Nasdaq-100 to a thirteen-month sample of 2,449 unscheduled news events from a public real-time newsfeed, with each event independently labelled by a large language model and benchmarked against the standard Loughran-McDonald dictionary and the FinBERT-tone neural classifier. After uniform Benjamini-Hochberg false discovery rate adjustment, the news events are robustly associated with intraday magnitude that is 1.3× to 1.9× the matched-baseline level on every asset-window cell tested, and this finding is stable under winsorisation, multivariate controls, and a pre/post-knowledge-cutoff split. The directional content of the LLM sentiment is modest (49–54% hit rate against realised direction) and below transaction costs in a backtest, but the LLM nonetheless outperforms the Loughran-McDonald baseline by 2–6 percentage points and the FinBERT baseline by up to 4.5 percentage points on the primary windows. The LLM is also the only one of the three classifiers capable of producing asset-specific sentiment (different USD and NDX labels on the same headline), which it does on 73.5% of events.

Two methodologically substantive findings sit alongside the main results. The pre-event drift in the fifteen minutes before each Discord headline is comparable in magnitude to the post-event drift, suggesting that the public publication time is not the informational event time in this setting; we recommend that future work on online-news event studies obtain primary-source timestamps where possible and report the pre-event drift as a diagnostic in any case. And the initial price reaction extends rather than reverses over the subsequent hours, with the sign of the +15-minute return agreeing with the sign of the +4-hour return on 57.6% of events and median absolute moves 3–4.5× larger over the longer window.

Two practical implications follow. For research, an LLM-based sentiment classifier provides meaningful value over the standard dictionary and BERT-family baselines in this kind of analysis, both in hit rate and in the asset-specific expressiveness that single-output classifiers cannot match; the full validation of the LLM labels nonetheless requires manual annotation, which we are pursuing. For practitioners using this kind of feed, the magnitude signal is more useful than the directional signal: news events are most reliably understood as triggers for volatility expansion, with direction available only as a weak secondary input that does not overcome execution costs in the naive forms tested.

We make the full pipeline—including the validation harness, the LLM sentiment cache, the dictionary and FinBERT baselines, and the three-way comparison—publicly available in a single repository with seeded reproducibility, in the hope that the methodology can be re-applied to other public newsfeeds and to a wider set of assets.

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A Reproducibility Statement

All numerical results in this paper can be reproduced from the public project repository using the commands and the seed reported below.

Code, data, and environment

The repository is organised around eight pipeline scripts plus auxiliary tools. Production runs use Python 3.13 in an isolated virtual environment with the dependencies pinned in `requirements.txt`. The event-study core uses `pandas`, `numpy`, `scipy`, `statsmodels`, and `matplotlib`; the LLM sentiment step uses the `openai` Python client against a DeepSeek API endpoint; and the dictionary baseline uses `pysentiment2`, which bundles the Loughran-McDonald master dictionary (Loughran and McDonald 2011). The FinBERT baseline uses `transformers` (4.x) with the `yiyanghkust/finbert-tone` pre-trained model (Yang et al. 2020).

Raw price data are downloaded via `dukascopy-python` (one-minute OHLCV bars for EUR/USD and the `E_NQ-100` CFD). Raw news data are exported via DiscordChatExporter from the publicly accessible FinancialJuice newsfeed channel and stored as a single JSON archive under `data/`. Neither input file is committed to the repository; both are reproducible from the underlying public sources.

Random seeds

All stochastic procedures (matched baseline sampling, bootstrap resampling, train/test splits used in robustness checks) are seeded with `SEED = 42` in `event_study.py`. Rerunning the pipeline with the same code, same input data, and the same seed reproduces every numerical value reported in Section 5 to the full precision of the output CSVs.

End-to-end reproduction

The full pipeline is reproduced by the following sequence of commands. Steps 1–3 download or generate the inputs; steps 4–9 produce the analysis outputs.

```
# 1. Parse the Discord export into the gold event list.
python parse_fj_discord.py "data/FinancialJuice ... .json" \
    -o outputs/events.csv --summary

# 2. Download or refresh the 1-minute price panels.
python download_prices.py --start 2025-03-24 --merge-existing

# 3. Classify each gold event with the LLM (cached in SQLite).
python sentiment.py outputs/events.csv \
    -o outputs/events_sentiment.csv --workers 15

# 4. Run the event-study pipeline (H1-H14, C1-C7).
python event_study.py

# 5. Validate methodology-sensitive outputs.
python validate_outputs.py
# Expected output: "OK: outputs passed methodology sanity checks"

# 6. Generate the HTML report.
python make_report.py --also-copy docs/report.html

# 7. Run the Loughran-McDonald dictionary baseline.
python sentiment_baseline.py outputs/events_sentiment.csv \
    -o outputs/events_sentiment_baseline.csv

# 8. Run the FinBERT-tone neural baseline (first run downloads
#    the yiyanghkust/finbert-tone model, ~440 MB, into HF cache).
python sentiment_finbert.py outputs/events_sentiment.csv \
```

```
    -o outputs/events_sentiment_finbert.csv

# 9. Three-way comparison on event sample: LLM vs LM vs FinBERT.
python sentiment_baseline_compare.py

# 10. External benchmark validation on Financial PhraseBank
#     (cached in outputs/sentiment_external_cache.sqlite).
python sentiment_external_benchmark.py
```

The manual-validation subsample for the sentiment audit is generated by `prepare_manual_validatio` and scored, once the two-annotator labels are filled in, by `score_manual_validation.py`.

Validation harness

`validate_outputs.py` runs deterministic sanity checks on the generated outputs: it verifies that the event/baseline magnitude ratio in `h1_results.csv` is between 1.0 and 5.0; that the H2 hit-rate column is in $[0, 1]$; that `methodology_summary.csv` reports the expected target conventions; and that the cluster-level outputs are non-empty. A pipeline run with code changes that breaks any of these invariants will fail this check before any results are reported.

Pre-specification

The fourteen pre-specified hypotheses H1–H14 are defined in `event_study.py` and were fixed before any cleanup of the result tables. The extensions C1–C7 were added during methodological revision, as documented in `STATUS.md`; C8 was added as an exploratory analysis and is reported in Section 7 rather than as a main result. Robust standard errors, cluster-robust standard errors, and the Benjamini-Hochberg false discovery rate adjustment (Benjamini and Hochberg 1995) are applied uniformly to every test output and are not chosen post hoc per hypothesis.

Repository

The complete source for the analysis and this paper draft is maintained as a git repository. Each subsection in Section 5 corresponds to a CSV file in `outputs/`; readers seeking to audit a specific number reported in the text can locate the underlying row by the variable name conventions described in `validate_outputs.py`.